Department of Mechanical and Aerospace Engineering

Brunel University of London

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Module Title:	Advanced Materials & Manufacturing
Module Tutor:	**Professor Hari Nadendla**
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Abstract

This study investigates how melt treatment, solidification behaviour, and heat treatment influence the properties of cast aluminium alloys. Melt quality was assessed using density index (DI), while cooling curve analysis evaluated solidification with and without Nb-based grain refinement. Mechanical properties were obtained from tensile testing of HPDC samples under different heat treatments. Results show that increasing DI reduces ductility and thermal conductivity due to porosity, while Nb refinement decreases undercooling and refines grain structure. Heat treatment significantly affects performance. Overall, the results highlight the strong interplay between melt quality, microstructure, and processing route.

Introduction

Aluminium alloys are widely used in engineering applications due to their low density, high specific strength, and good castability, making them suitable for automotive, aerospace, and structural components [1]. However, their performance is highly dependent on processing conditions, particularly melt quality [2], solidification behaviour, and microstructural control. Defects such as porosity can significantly reduce mechanical properties [3], while grain refinement and heat treatment are commonly used to improve strength and reliability. This set of experiments is important as it replicates key industrial processes, including degassing, grain refinement, permanent mould casting, and high pressure die casting (HPDC). Understanding how these processes influence microstructure and mechanical properties is critical for designing high-performance aluminium components. The experiments provide insight into the relationship between processing parameters, defect formation, and final material behaviour.

The Experimental Problem & Theory

The objective of this study is to establish a quantitative relationship between processing conditions, microstructural evolution, and the resulting mechanical and physical properties of Al-Si based casting alloys. Specifically, the work examines how melt quality, solidification behaviour, and post-solidification heat treatment determine porosity formation, grain structure, and tensile performance. In cast aluminium systems, the dominant defect mechanism is hydrogen-induced porosity. Hydrogen exhibits high solubility in liquid aluminium but negligible solubility in the solid phase [4]. The tendency of a melt to form such porosity is quantified using the DI obtained from reduced pressure testing, shown in Equation 1:

$$DI = \frac{D_{atm} - D_{vac}}{D_{atm}} \times 100 \qquad \text{Equation 1}$$

where D_{atm} and D_{vac} are the densities of samples solidified under atmospheric and reduced pressure, respectively. Reduced pressure promotes pore growth, amplifying differences in porosity. Therefore, DI represents a measure of melt integrity, where higher values indicate greater hydrogen content, increased porosity formation, and reduced structural reliability.

Solidification behaviour is analysed using cooling curves to identify phase transformations and nucleation characteristics. The addition of Al-Nb-B grain refiners introduces heterogeneous nucleation sites, lowering the nucleation barrier compared to homogeneous nucleation [5]. This reduces undercooling, suppresses recalescence, and promotes finer, equiaxed grain structures, improving mechanical uniformity and reducing defect sensitivity. The solidification path of the Al-10Si-based Aural alloy follows a hypoeutectic sequence which is primary α-Al nucleates first, followed by the formation of an α-Al + Si eutectic, with minor intermetallic phases forming from residual solute enrichment [6]. The eutectic fraction can be estimated using the lever rule applied to the Al–Si binary phase diagram, providing a first-order approximation of phase proportions.

Mechanical behaviour is characterised using tensile testing, where engineering stress and strain are defined as:

$$\sigma = \frac{F}{A_0}, \varepsilon = \frac{\Delta L}{L_0} \qquad \text{Equation 2}$$

where $F$is the applied load, $A_0$is the original cross-sectional area, and $L_0$is the initial gauge length. The 0.2% proof stress is determined using an offset method to define the onset of plastic deformation in materials without a clear yield point. The ultimate tensile strength (UTS) represents the maximum load-bearing capacity, while ductility to fracture reflects the material's ability to accommodate plastic deformation prior to failure. Heat treatment modifies the microstructure through diffusion-controlled processes. Solution treatment dissolves segregated phases and homogenises the microstructure, reducing strength but increasing ductility [7]. Subsequent artificial ageing promotes the formation of fine, coherent precipitates, which impede dislocation motion and increase strength. Over-ageing results in precipitate coarsening and loss of coherency, reducing strengthening effectiveness. In HPDC alloys, these effects are strongly coupled with pre-existing porosity, which acts as stress concentrators and limits the achievable mechanical performance.

Methodology

The methodology was designed to isolate the effects of melt quality, solidification behaviour, and post-processing on alloy performance, and was divided into three stages: melt treatment and casting, solidification analysis, and mechanical testing.

In Section A, melt quality was controlled via rotary degassing by varying gas flow rate, degassing time, and post-degassing hold time, which govern hydrogen removal and reabsorption. Melt integrity was quantified using the reduced pressure test to calculate DI. Tensile properties and thermal conductivity were measured on gravity die cast specimens to assess the impact of porosity on mechanical and physical performance. In Section B, solidification behaviour was analysed using cooling curves for Aural alloy with and without 0.1 wt.% Nb (Al-Nb-B). Temperature–time data and first-derivative analysis were used to identify liquidus, eutectic, and solidus temperatures. Microstructures were examined using optical microscopy, and grain size was quantified using the linear intercept method. Grain size evolution with holding time was also analysed to assess grain refiner fading behaviour. In Section C, tensile testing was performed on HPDC samples in as-cast, solution-treated, T6, and over-aged conditions. Load–extension data were converted into engineering stress–strain curves using specimen geometry, and 0.2% proof stress, UTS, and ductility were extracted. Results were averaged over three tests to ensure reliability, allowing evaluation of the combined effects of rapid solidification, porosity, and heat treatment.

Section A

The integrity of cast aluminium alloys is governed by melt hydrogen content, as hydrogen is highly soluble in the liquid but not in the solid, leading to pore formation during solidification. The density index (DI), obtained from reduced pressure testing, is therefore a practical measure of melt soundness and porosity susceptibility: low DI indicates a clean melt, while high DI indicates increased porosity, reduced load-bearing area, and earlier failure.

DI is controlled by the balance between hydrogen pickup and removal, with key factors including melt temperature, humidity, alloy composition, degassing efficiency, and post-degassing hold time. While higher gas flow rates and degassing times improve hydrogen removal, the melt can rapidly reabsorb hydrogen during holding. As shown in Figure 1, DI increases sharply with hold time, rising from 0.4 at 0 min to 13.1 at 240 min, demonstrating that post-degassing re-equilibration is the dominant factor degrading melt quality.

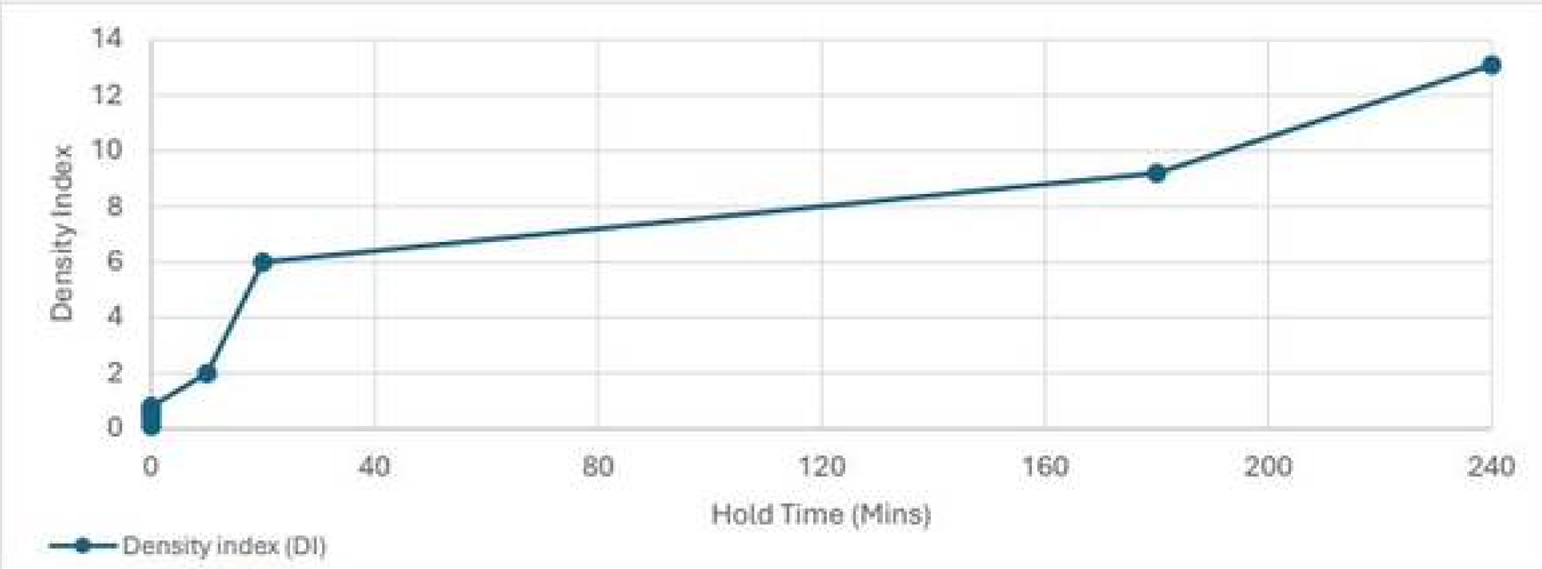

Figure 1: Increase in density index with post-degassing hold time for cast Aural alloy.

The measured tensile properties show that increasing DI reduces mechanical performance, although the sensitivity differs between properties, as shown in Figure 2 and Figure 3:

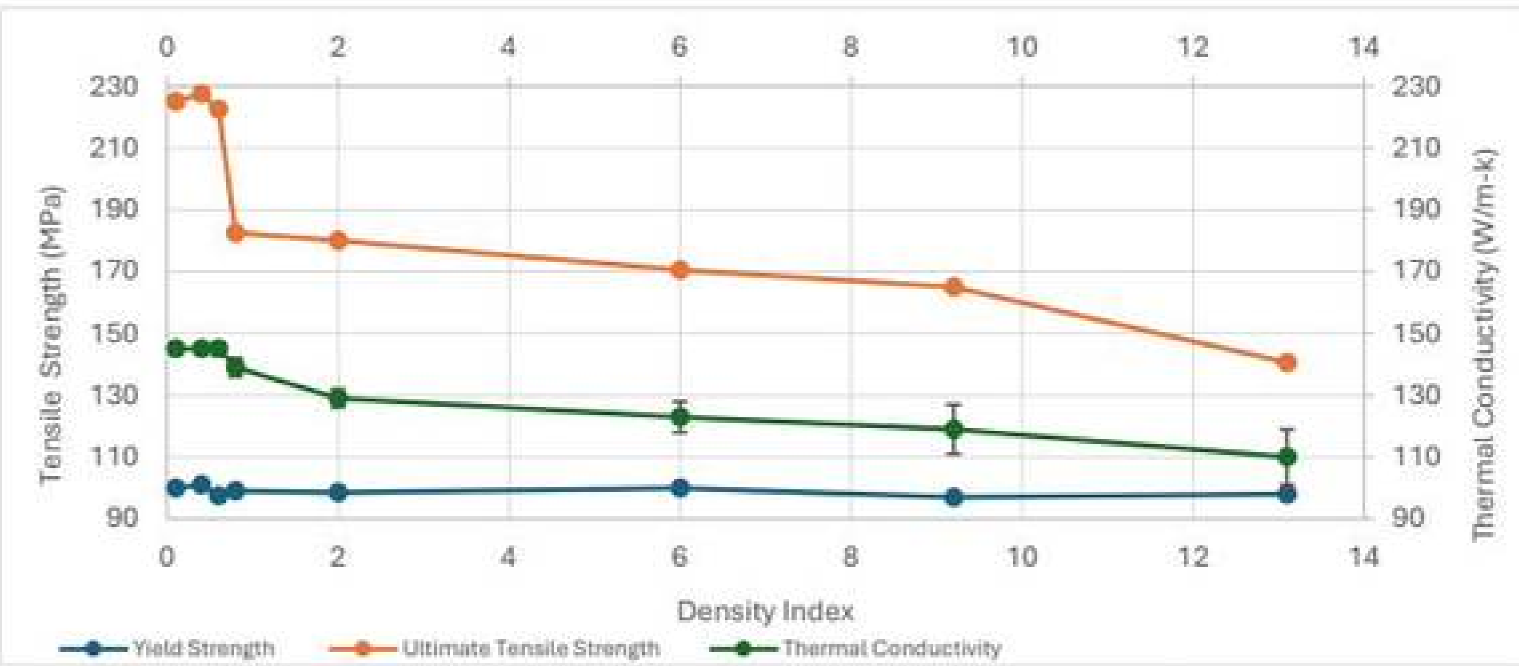

Figure 2: Effect of density index on yield strength, UTS, and thermal conductivity of cast Aural alloy.

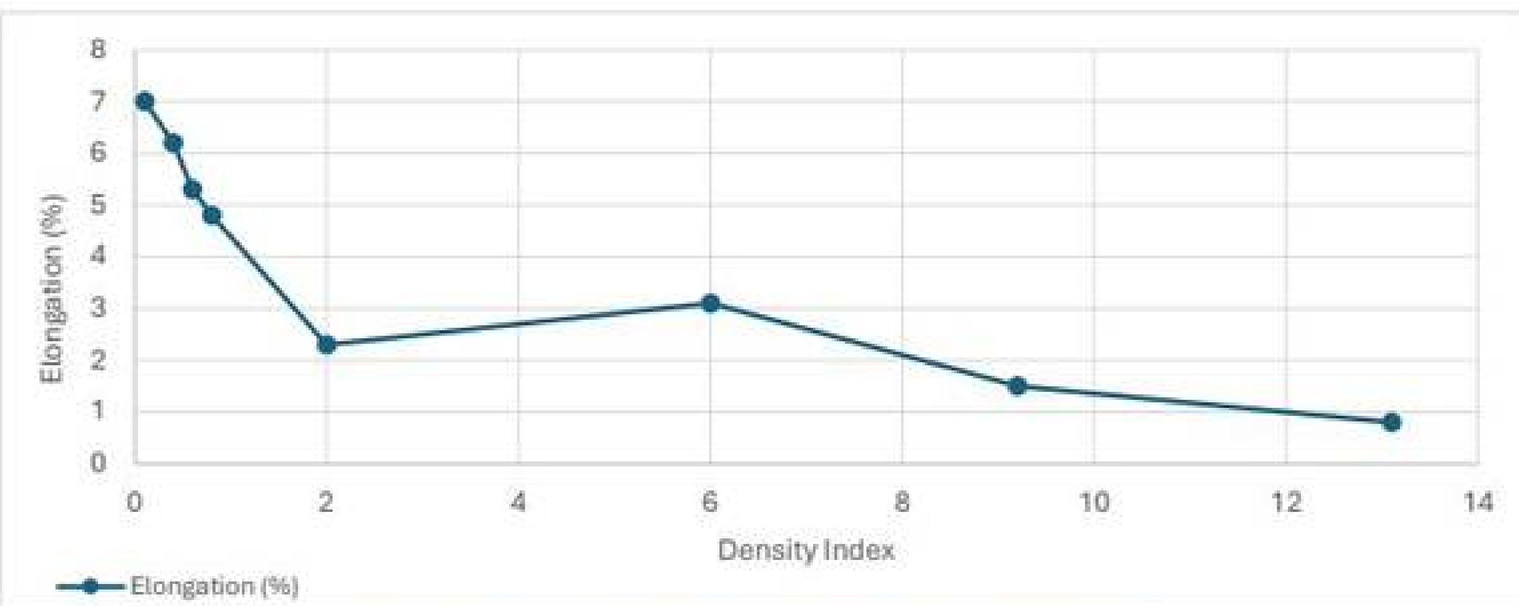

Figure 3: Effect of density index on elongation of cast Aural alloy.

Yield strength remains nearly constant across the full DI range, varying only from about 97 to 101 MPa. This suggests that yielding is controlled mainly by the alloy matrix and is relatively insensitive to the porosity levels generated here. In contrast, UTS decreases substantially with increasing DI, falling from approximately 225–227.5 MPa at low DI to 140.5 MPa at DI = 13.1. This reduction occurs because pores act as internal defects that reduce the effective cross-sectional area and create local stress concentrations, causing earlier crack initiation and faster propagation under tensile loading. Elongation shows the greatest sensitivity to DI and is therefore the property most strongly affected by casting defects. As shown in Figure 3, elongation falls sharply from 7.0% at DI = 0.1 to 0.8% at DI = 13.1, confirming that ductility is highly defect-sensitive. This behaviour is characteristic of pore-controlled fracture: pores localise plastic strain, accelerate void coalescence, and drastically reduce the ability of the material to sustain plastic deformation before failure. Although there is a small local increase in elongation at intermediate DI, the overall trend is negative and consistent with deterioration in casting quality.

Thermal conductivity also decreases systematically as DI increases, as shown in Figure 2. At low DI values of 0.1–0.6, the conductivity remains close to 145 W/m·K, whereas at DI = 13.1 it falls to approximately 110 W/m·K. This decline occurs because porosity interrupts continuous metallic heat-flow paths and reduces the effective conducting area within the casting. In addition, the larger error bars at high DI indicate increasing variability in conductivity, which is consistent with a less uniform and more heterogeneous pore distribution in poorer-quality melts. The conductivity results therefore support the tensile data and confirm that increasing DI corresponds to a progressive loss of both structural and functional performance.

Some uncertainty is associated with the results due to variability in measurements and pore distribution, particularly at higher DI where error increases. Additionally, fluctuations in melt handling and exposure time may introduce variability in hydrogen reabsorption, affecting the measured DI. Overall, the results demonstrate that the DI is a robust indicator of casting integrity in Aural alloy. Increasing DI has little effect on yield strength but causes major reductions in UTS, ductility, and thermal conductivity due to hydrogen-induced porosity.

Section B

The cooling curves for the Aural alloy with and without Al-Nb-B grain refiner show the key thermal events associated with solidification and clearly demonstrate the effect of heterogeneous nucleation on the solidification path. As shown in Figure 4, both alloys initially undergo rapid cooling in the liquid state, followed by a deviation from linear cooling at the onset of primary solidification. The liquidus temperature is approximately 590°C for the reference alloy and 593°C for the Nb-refined alloy. The reference alloy then exhibits clear undercooling to approximately 586°C, followed by recalescence, indicating that nucleation is delayed until a sufficient thermodynamic driving force has developed. In contrast, the Nb-refined alloy shows only minimal undercooling and a much smoother transition into solidification, indicating that the grain refiner promotes earlier nucleation and reduces the need for undercooling.

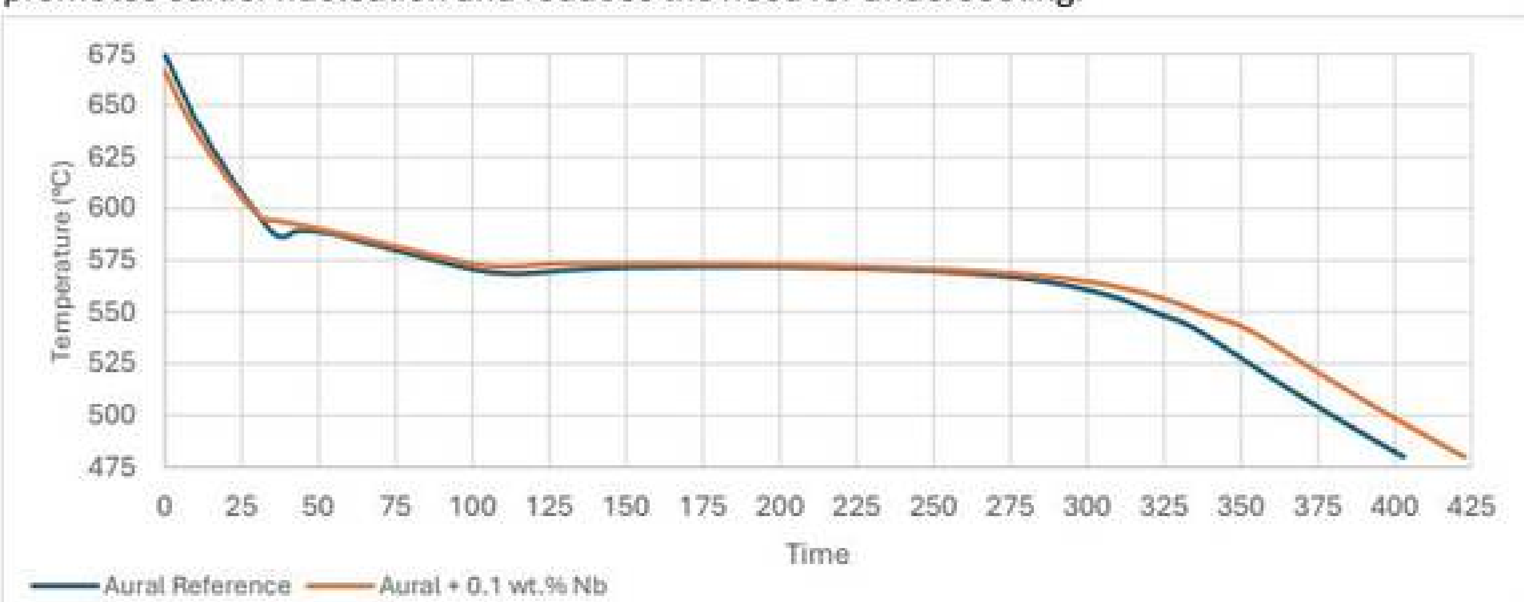

Figure 4: Cooling curves for permanent mould cast Aural alloy with and without 0.1 wt.% Nb grain refiner addition.

At later times, both curves show a near-isothermal region associated with the eutectic reaction. The eutectic temperature is approximately 571°C for the reference alloy and 573°C for the Nb-refined alloy. The end of this thermal arrest marks completion of solidification, giving an estimated solidus temperature of 569°C for the reference alloy and 571°C for the Nb-refined alloy. These results confirm that both alloys follow the same overall solidification sequence, but the addition of Nb significantly modifies the early stages of solidification by altering nucleation behaviour rather than changing the fundamental phase evolution. The key temperatures extracted from the cooling curves are summarised in Table 1:

Table 1: Key temperatures extracted from cooling curves for Aural alloy with and without 0.1 wt.% Nb

Property	**Reference Alloy**	**Aural + 0.1 wt.% Nb**
Liquidus Temperature	590°C	593°C
Undercooling Minimum	586°C	593°C (Minimal Undercooling)
Eutectic Temperature	571 °C	573°C
Solidus Temperature	569°C	571 °C

The differences in cooling curve shape are a direct consequence of the effect of heterogeneous nuclei introduced by the Al-Nb-B master alloy. In the unrefined alloy, nucleation must occur on a relatively limited number of naturally available substrates, so a greater degree of undercooling is required before primary α-Al can form. This leads to the pronounced recalescence peak seen in the reference cooling curve. With Nb addition, potent inoculant particles provide favourable nucleation substrates, lowering the effective nucleation barrier. Compared with homogeneous nucleation, which would require very large undercooling and is not favoured under normal casting conditions, heterogeneous nucleation enables solidification to begin earlier and more uniformly throughout the melt. As a result, undercooling is reduced, recalescence is suppressed, and the solidification process becomes more thermally stable. This behaviour shows that Nb-B inoculants increase nucleation efficiency and promote a finer, more equiaxed grain structure.

The solidification path of Aural alloy is consistent with that of a hypoeutectic Al-Si alloy. On cooling from the liquid state, primary α-Al forms first at the liquidus. As solidification proceeds, the remaining liquid becomes enriched in silicon until the eutectic composition is reached, at which point the remaining liquid transforms through the eutectic reaction to form α-Al + Si. At room temperature, the microstructure therefore consists primarily of primary α-Al, an Al-Si eutectic constituent, and minor intermetallic phases associated with alloying elements such as Fe and Mn. Using the simplified Al-10Si binary approximation, the eutectic fraction can be estimated from the lever rule, shown in Equation 3:

$$f_{eutectic} = \frac{C_0 - C_{\alpha e}}{C_e - C_{\alpha e}} = \frac{10 - 1.65}{12.6 - 1.65} = 0.763 \qquad \textit{Equation 3}$$

Thus, the alloy contains approximately 76.3% eutectic and 23.7% primary α-Al. This high eutectic fraction is consistent with the observed extended eutectic arrest in the cooling curve and the characteristic mixed microstructure seen in the metallographic images.

Grain size measurements using the linear intercept method confirm significant refinement with Nb addition. Using a test line length of 5645 µm, the unrefined alloy produced 7 intercepts (~800 µm), while the Nb-refined alloy produced 22 intercepts (~260 µm). The microstructures in Figure 5A and 5B support this, showing coarse grains in the reference alloy and a finer, more uniform structure with Nb. This refinement results from an increased density of heterogeneous nucleation sites, which promotes more nucleation events and limits grain growth. Likely inoculant phases include NbB_2 and Nb-rich aluminides, which act as effective nucleation substrates for α-Al.

Figure 5A (Left): Optical microstructure of unrefined Aural alloy (30 mm diameter PMC sample, as-cast condition). Figure 5B (Right): Optical microstructure of Aural alloy containing 0.1 wt.% Nb added as Al-Nb-B master alloy (30 mm diameter PMC sample, as-cast condition).

The fading behaviour of the Al-Nb-B grain refiner is shown in Figure 6, where grain size increases with holding time after grain refiner addition. Immediately after treatment, the grain size falls sharply to around 200–250 µm, confirming strong initial refinement. However, with increasing holding time the grain size progressively increases, particularly in the non-stirred condition, showing that refinement efficiency decreases with time. The stirred melts consistently retain smaller grain sizes than the non-stirred melts at longer holding times, indicating that agitation helps maintain a more uniform suspension of active inoculant particles and delays the loss of refinement efficiency. This fading behaviour is attributed to a progressive reduction in the number of effective nucleation sites, which may arise from particle settling, agglomeration, dissolution, or interfacial deactivation during holding. To reduce the risk of fading in practice, the grain refiner should be added as close as possible to casting, long holding times after addition should be avoided, and melt stirring should be maintained where possible to preserve a uniform distribution of active particles.

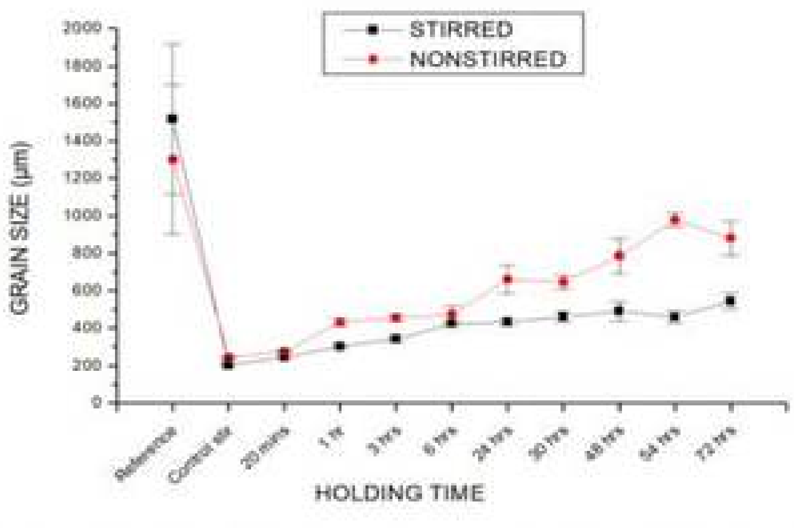

Figure 6: Fading behaviour of the grain refiner in the alloy showing grain size variation for stirred and non-stirred melts.

Uncertainty in grain size measurements arises from limitations of the linear intercept method, including line placement, image resolution, and the limited number of grains sampled. Similarly, identification of transformation temperatures from cooling curves is subject to minor error due to noise in the thermal data and interpretation of inflection points. Overall, the results show that Al-Nb-B grain refinement significantly changes the thermal and microstructural response of Aural alloy during permanent mould casting. The grain refiner reduces undercooling, suppresses recalescence, refines grain size from approximately 800 µm to 260 µm,

Section C

The primary source of porosity in HPDC is gas entrapment during high-speed, turbulent die filling, where air is entrained and trapped during solidification. Unlike permanent mould casting, where porosity is mainly hydrogen-related, HPDC porosity is predominantly process-induced. The addition of Al-Nb-B grain refiner does not eliminate porosity but refines its distribution; a finer grain structure promotes a more uniform solidification front, reducing defect localisation and mitigating its impact on tensile performance. Tensile behaviour was determined by converting load–extension data into engineering stress and strain using $A_0 = 30.19\,\text{mm}^2$and $L_0 = 50\,\text{mm}$. The resulting stress–strain curves shown in Figure 7:

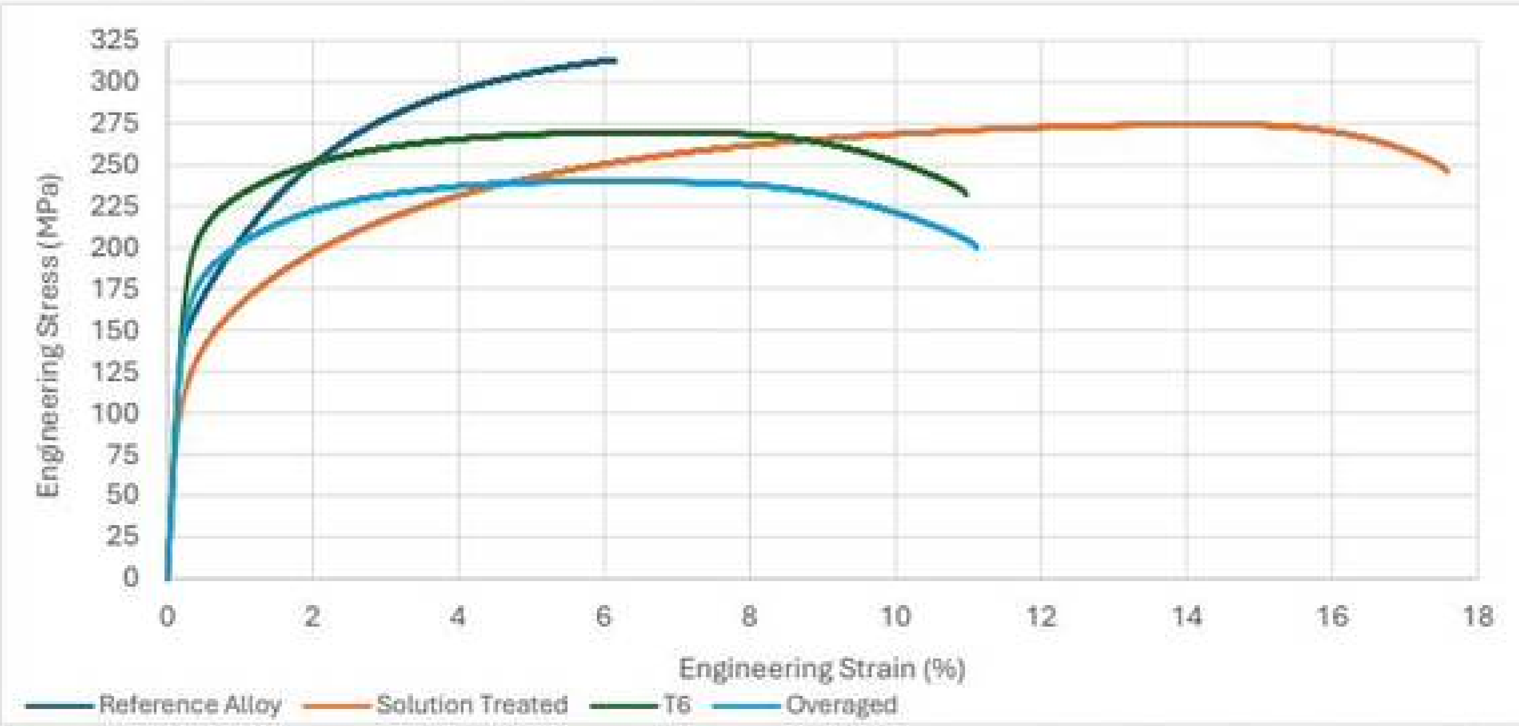

Figure 7: Engineering stress-strain curves for Aural alloy under as-cast, solution-treated, T6, and over-aged conditions.

The tensile properties extracted from the curves are summarised in Table 2:

Table 2: Tensile properties of HPDC Aural alloy under different heat-treatment conditions

Condition	0.2% Proof Stress (MPa)	UTS (MPa)	Ductility To Fracture (%)
As-Cast	168	317.3	6.9
Solution	127	272.9	14.4
T6	206	265.3	6.4
Over-Aged	171	232.7	6.2

The as-cast condition shows a 0.2% proof stress of 168.3 MPa, UTS of 317.3 MPa, and ductility of 6.9%. Solution treatment reduces strength (127.3 MPa, 272.9 MPa) while significantly increasing ductility to 14.4% due to dissolution of strengthening phases and microstructural homogenisation, which lowers stress concentrations and allows greater plastic deformation. The T6 condition provides the highest proof stress (206 MPa) with a UTS of 265.3 MPa and ductility of 6.4%, as fine precipitates formed during ageing impede dislocation motion. In contrast, the over-aged condition shows reduced strength (170.7 MPa, 233 MPa) with low ductility (6.2%) due to precipitate coarsening and loss of strengthening effectiveness. Overall, heat treatment governs the strength–ductility balance: solution treatment maximises ductility, T6 maximises strength, and over-aging reduces strengthening performance.

A notable result is that the as-cast condition exhibits the highest UTS, even though it is not the strongest condition in terms of proof stress. This indicates that tensile performance in HPDC alloys is not governed by heat treatment alone, but also by the interaction between strengthening mechanisms and pre-existing defects. In HPDC material, entrapped gas pores act as crack initiation sites and can limit the benefit of elevated-temperature heat treatment. During solution treatment and subsequent ageing, these pores may become more detrimental to tensile performance, reducing the achievable UTS even when the matrix itself is strengthened. This explains why the T6 condition has the highest proof stress but does not exceed the as-cast condition in UTS. A comparison of the results is best shown in Figure 8:

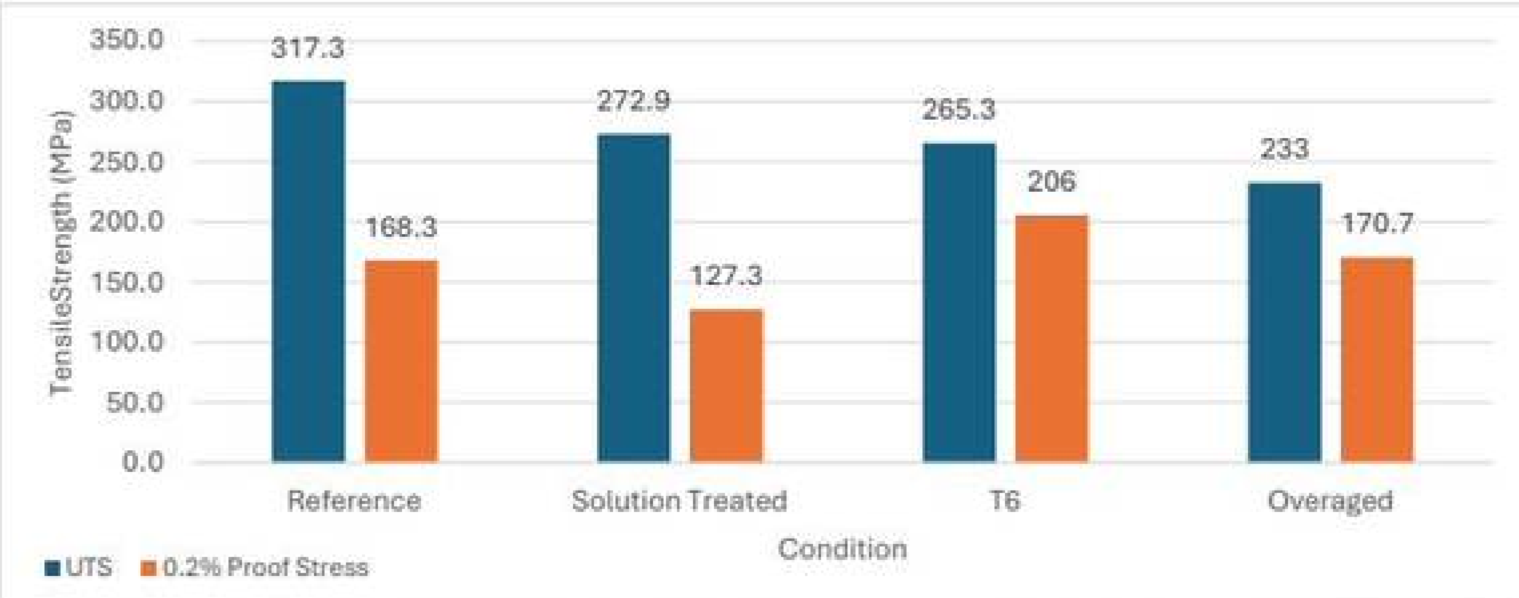

Figure 8: Comparison of ultimate tensile strength and 0.2% proof stress for HPDC Aural alloy under different heat-treatment conditions.

Comparison with the permanent mould cast samples from Section A highlights the strong influence of processing route. The permanent mould samples exhibit lower tensile strengths overall, with UTS values ranging from approximately 140.5 to 227.5 MPa depending on , and elongation ranging from 0.8 to 7.0%. In contrast, the HPDC samples show substantially higher strength, with UTS values up to 317.3 MPa, and ductility that is comparable or significantly higher in the solution-treated condition. The main reason for this difference is the much higher cooling rate in HPDC, which produces a finer microstructure and therefore higher strength. However, the comparison also shows the limitation of HPDC: although rapid solidification improves strength, the process is more sensitive to gas entrapment, so porosity remains a major constraint on ductility and tensile reliability.

Variability in tensile results may arise from specimen-to-specimen differences in porosity distribution, as well as measurement uncertainty in extensometer readings and specimen dimensions. In particular, the determination of 0.2% proof stress is sensitive to the offset method and curve interpretation. Overall, the results show that the tensile properties of HPDC Aural alloy are controlled by the combined effects of solidification rate, heat treatment, and defect population. Heat treatment changes the balance between strength and ductility through precipitation and microstructural evolution, but the final mechanical behaviour remains strongly limited by gas porosity introduced during HPDC processing. For this reason, optimisation of HPDC alloys requires not only appropriate heat treatment selection, but also strict control of melt handling and die filling in order to minimise defect formation.

<u>**References**</u>

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